

Affiliate Network Integration Design

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Overview

This document describes how LYKE integrates with an affiliate network to collect commission revenue from retailers. It covers commercial setup, technical tracking changes, conversion ingestion, transaction reconciliation, and the downstream impact on creator payments.

The core insight is that affiliate networks solve the "who pays LYKE?" problem by acting as a trusted financial intermediary between retailers and publishers. LYKE never needs to invoice retailers or build billing infrastructure for affiliate revenue. The network validates purchases, deducts its fee, and pays LYKE monthly.

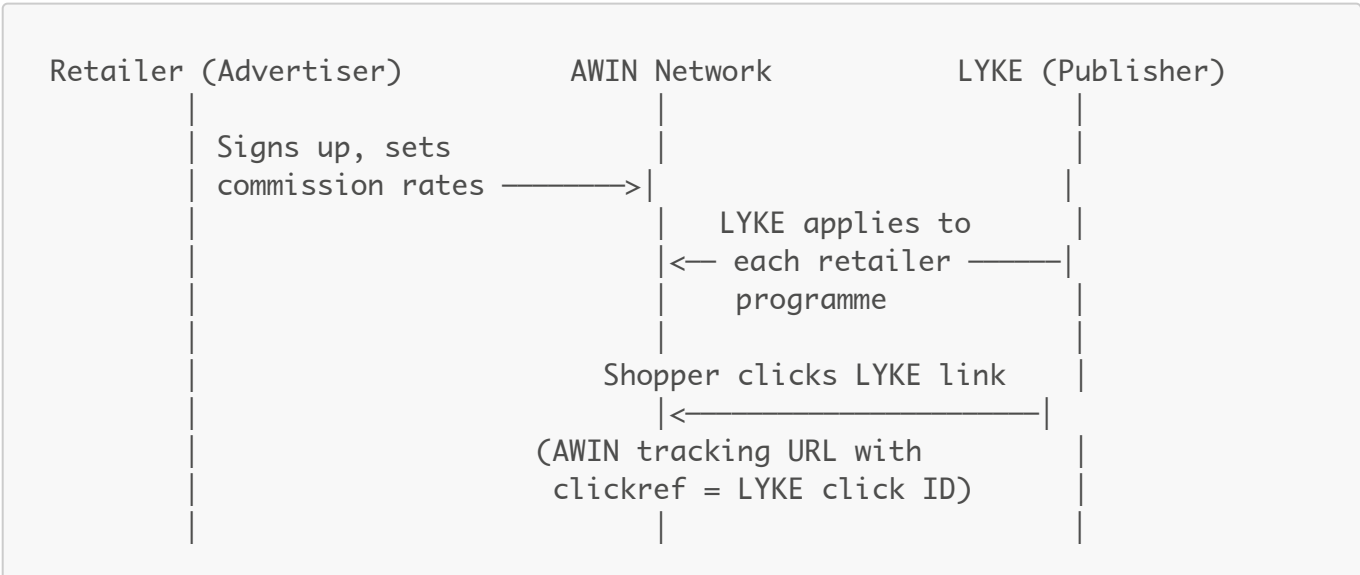
Recommended Network: AWIN

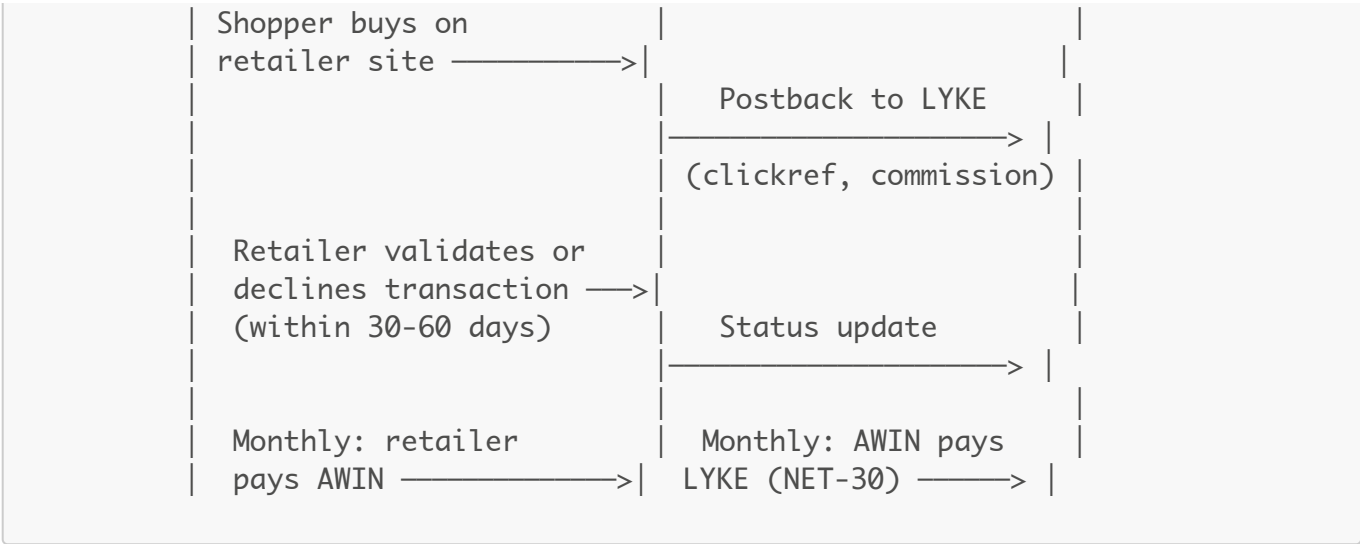
AWIN (formerly Affiliate Window) is the recommended first integration for three reasons:

- 1. **UK fashion depth** — AWIN hosts the affiliate programmes of most major UK fashion retailers: ASOS, M&S, Reiss, Phase Eight, White Stuff, Jigsaw, and many others. A single AWIN publisher account gives LYKE access to all of them.
- 2. **Fit with existing architecture** — the `AffiliateConfig` JSON already stored on `Retailer` maps almost exactly to AWIN's link format. No data model rethink is required for link generation.
- 3. **Server-to-server postback** — AWIN supports a postback URL (their equivalent of a conversion webhook) that LYKE can configure once per account, then never touch again.

Other networks worth evaluating later: **Rakuten Advertising** (strong in premium/luxury), **Impact** (good API ergonomics), **Partnerize** (enterprise). All use the same publisher/advertiser model and largely interchangeable concepts.

How Affiliate Networks Work





Terminology:

Term	AWIN Name	LYKE equivalent
LYKE	Publisher	Publisher account
Retailer	Advertiser	Retailer entity
Conversion tracking link	AWIN tracking URL	Generated by <code>GenerateAffiliateUrl()</code>
Click identifier	<code>clickref</code>	<code>ClickEvent.Id</code> (GUID)
Conversion notification	Server-to-server postback	Conversion webhook
Commission validation	Transaction validation	<code>Pending</code> → <code>Confirmed</code> transition
Commission reversal	Transaction declined	Currently unmodelled
AWIN's cut	Network override	~3% of transaction value (varies)

Part 1: Commercial Setup (Outside of Code)

These are the steps LYKE takes once, commercially, before any code changes are needed.

Step 1 — Register as an AWIN Publisher

- 1. Apply at `ui.awin.com` as a Publisher (website/app)
- 2. Describe LYKE: mobile fashion app, body-profile matching, referral via in-app product links
- 3. AWIN assigns a **Publisher ID** (e.g., `pub_123456`) — this is stored in config, not per-retailer
- 4. AWIN will request a brief review of the app/concept before approval (typically 1-5 days)
- 5. Once approved, access to the AWIN publisher dashboard and API is granted

Step 2 — Apply to Retailer Programmes

For each retailer LYKE wants to work with:

- 1. Find the retailer's programme in the AWIN marketplace
- 2. Submit an application (typically a short paragraph explaining the referral method)
- 3. The retailer reviews and approves (days to weeks; established retailers are faster)

4. Once approved, the retailer's **Merchant ID** (e.g., `mid_7890`) becomes available
5. Record the Merchant ID in LYKE's `Retailer.AffiliateConfig` JSON

Retailers set their own commission rates within AWIN (e.g., 8% on apparel, 5% on footwear). AWIN's standard publisher share of that is approximately 75-80% (they retain 20-25% as a network override fee). The `CommissionAmount` AWIN reports in postbacks and the API is the **publisher's net amount** — i.e., what LYKE actually receives after AWIN's cut. No calculation is needed.

Step 3 — Configure Postback URL in AWIN Dashboard

AWIN supports server-to-server (S2S) postbacks for conversion tracking. This replaces the current per-retailer conversion webhook.

Register the following postback URL once in the AWIN publisher dashboard:

```
https://api.lyke.app/api/webhooks/awin?  
transaction_id={transaction_id}  
&order_ref={order_ref}  
&pence={pence}  
&clickref={clickref}  
&merchant_id={merchant}  
&status={status}
```

AWIN will call this URL for every conversion attributed to LYKE. The `{clickref}` will contain LYKE's `ClickEvent.Id` GUID, enabling end-to-end attribution.

Step 4 — Agree Payment Terms with AWIN

AWIN pays publishers monthly, typically NET-30 (payment issued 30 days after the close of the month). Minimum payment threshold (usually £20 for bank transfer, £0 for BACS). Payment is via bank transfer.

This payment schedule has a direct bearing on creator payment timing — see Part 4.

Part 2: Tracking Link Changes

How Links Are Built Today

`GenerateAffiliateUrl()` in `CommerceService.cs` reads `Retailer.AffiliateConfig` (JSON) and assembles:

```
{BaseUrl}?{SubIdParam}={clickId}&{ProductUrlParam}={encodedProductUrl}&  
{customParams}
```

For a direct retailer integration the `BaseUrl` would be something like `https://retailer.com/track`.

How Links Work with AWIN

AWIN's canonical tracking URL format is:

```
https://www.awin1.com/cread.php
?awinmid={merchant_id}
&awinpid={publisher_id}
&clickref={click_id}
&p={encoded_product_url}
```

The existing `AffiliateConfig` JSON fields map directly:

```
{
  "Enabled": true,
  "NetworkType": "AWIN", // new
  field
  "BaseUrl": "https://www.awin1.com/cread.php?awinmid=7890&awinpid=123456",
  "Parameters": {
    "SubIdParam": "clickref", //
    LYKE's click ID goes here
    "ProductUrlParam": "p"
  },
  "CommissionRate": 0.08, //
  informational only; AWIN reports actual
  "CookieWindowDays": 30,
  "WebhookSecret": null // not
  used with AWIN (postback is unsigned)
}
```

No changes are needed to `GenerateAffiliateUrl()`. Once each retailer's `AffiliateConfig` is populated with the correct AWIN `BaseUrl` (containing the Merchant ID and Publisher ID), link generation works immediately.

The LYKE click ID (a GUID) maps to AWIN's `clickref` parameter, which AWIN stores and echoes back in the postback. This is the attribution chain.

Mobile Deep Link Consideration

AWIN's standard `cread.php` redirect works in a browser but can behave unpredictably in an in-app browser (Capacitor Browser). AWIN provides a **mobile-optimised link format** via their API:

```
https://www.awin1.com/awclick.php?mid={merchant_id}&id=
{publisher_id}&clickref={click_id}&p={encoded_product_url}
```

`awclick.php` is AWIN's redirect endpoint designed for mobile contexts. It performs a direct 302 redirect without JavaScript. LYKE should use this format for all mobile links. The `BaseUrl` in `AffiliateConfig` should use `awclick.php` rather than `cread.php`.

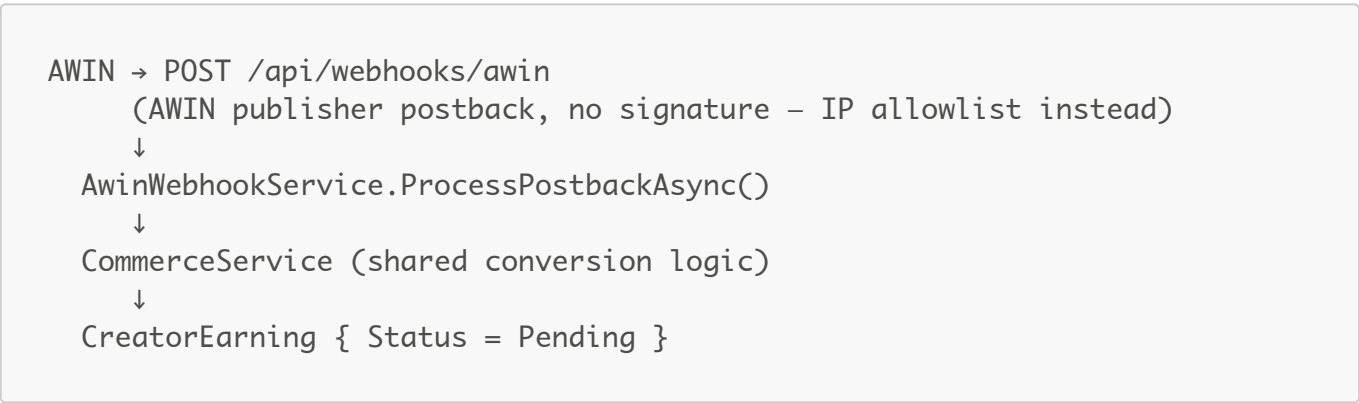
Part 3: Conversion Ingestion Changes

Current Design



This design works for direct retailer integrations but has problems for network integrations: the retailer never calls LYKE directly — AWIN does.

New Design



AWIN Postback Endpoint

AWIN postbacks are not signed with HMAC. Instead, AWIN sends postbacks from a small set of known IP ranges. The endpoint must:

- 1. Verify the caller's IP against AWIN's published IP allowlist (updated via their API or docs)
- 2. Map the `merchant_id` query parameter to a `Retailer` record
- 3. Map `clickref` to a `ClickEvent.Id` GUID
- 4. Map `pence` (integer, in the retailer's minor currency unit) to a `decimal` commission amount
- 5. Handle the `status` field (`pending` → create earning, `declined` → reverse earning)

AWIN postback query parameters:

Parameter	Type	Description
<code>transaction_id</code>	string	AWIN's internal transaction ID — store for reconciliation
<code>order_ref</code>	string	Retailer's order reference
<code>pence</code>	int	Commission in minor units (pence for GBP)
<code>clickref</code>	string	LYKE's <code>ClickEvent.Id</code> (GUID)
<code>merchant_id</code>	int	AWIN Merchant ID — maps to a <code>Retailer</code>

Parameter	Type	Description
status	string	pending or declined
currency	string	ISO 4217 currency of commission

Storing the AWIN Transaction ID

The `ClickEvent.AttributionData` JSON field should be extended to store AWIN's `transaction_id`. This is needed for:

- Reconciliation against AWIN's monthly payment report
- Looking up which `CreatorEarning` to reverse if a transaction is later declined
- Auditing disputed transactions

Add `awinTransactionId` to the `attributionData` anonymous object in `CommerceService.TrackClickAsync()` when an AWIN retailer is detected, and also when the postback is processed.

Existing Direct Retailer Webhook

The existing `POST /api/retailers/{id}/conversions` endpoint should be **retained** for non-AWIN retailers (direct integrations). The two routes are not mutually exclusive. A retailer's `AffiliateConfig.NetworkType` determines which path is active.

Part 4: Transaction Status Sync

This is the most significant architectural change. The current `EarningConfirmationJob` (timer trigger, time-based) assumes that any `Pending` earning older than 30 days is safe to confirm. With AWIN, that assumption is wrong in two directions:

- AWIN may validate a transaction earlier than 30 days (common for fast-validating retailers)
- AWIN may extend the validation window beyond 30 days for high-return categories
- AWIN may **decline** a transaction at any point during the validation window

The correct approach is to drive `CreatorEarning` status from AWIN's authoritative transaction status, not from elapsed time.

AWIN Transaction Statuses

AWIN Status	LYKE <code>EarningStatus</code>
pending	Pending
confirmed	Confirmed
declined	Reversed (new)
deleted	Reversed (new)

New Status: `EarningStatus.Reversed`

Add **Reversed = 3** to the **EarningStatus** enum. This represents a commission that was once **Pending** (or even **Confirmed**) but was subsequently declined by the retailer — typically because the order was returned or the transaction was fraudulent.

Reversed earnings are excluded from all balance calculations and cannot be included in payout requests.

Option A: Status Postback (Preferred for Latency)

AWIN sends a second postback when a transaction's status changes (e.g., from **pending** to **confirmed** or **declined**). The postback URL is the same endpoint as the initial conversion postback — the **status** field distinguishes them.

Handling:

- **status=pending** → create **CreatorEarning { Status = Pending }** (already done)
- **status=confirmed** → find **CreatorEarning** by **awinTransactionId**, set **Status = Confirmed**
- **status=declined** or **status=deleted** → find **CreatorEarning**, set **Status = Reversed**

This is sufficient for most cases. The postback is best-effort (AWIN does not guarantee delivery), so Option B is needed as a fallback.

Option B: AWIN API Reconciliation Job (Fallback/Audit)

AWIN provides a transactions API:

```
GET https://api.awin.com/publishers/{publisherId}/transactions/  
?startDate=2026-01-01T00:00:00  
&endDate=2026-01-31T23:59:59  
&status=confirmed  
Authorization: Bearer {awin_api_token}
```

A nightly reconciliation job should:

1. Query AWIN's API for all transactions modified in the past 48 hours
2. For each transaction, look up the matching **CreatorEarning** by **awinTransactionId**
3. If the AWIN status differs from LYKE's status, update LYKE's record
4. Log any LYKE earnings with no matching AWIN record (attribution gap)

This job replaces the existing **EarningConfirmationJob**. The time-based logic (**CreatedAt < now - 30 days** → **Confirmed**) is removed. Confirmation is driven entirely by AWIN.

Schedule: Daily at 06:00 UTC, implemented as an Azure Functions timer trigger (consistent with existing pattern in **Lyke.Functions**).

Part 5: Creator Payment Considerations

The affiliate network model introduces several constraints and timing dependencies that directly affect how creator payments work.

5.1 LYKE Can Only Pay Creators After AWIN Pays LYKE

This is the most important constraint. The current payout design (from [specs/payout-design.md](#)) allows creators to request a payout as soon as their earnings are **Confirmed**. But **Confirmed** now means "AWIN has validated the transaction" — not "LYKE has received the cash".

AWIN pays LYKE on NET-30 terms: payment is issued approximately 30 days after the end of the calendar month in which transactions were confirmed. For example:

- Transaction confirmed by AWIN: 15 January
- AWIN pays LYKE: ~28 February
- Creator can be paid: March

This means the earliest a creator can realistically receive payment is 6-10 weeks after a sale — the retailer's validation window (up to 30-45 days) plus AWIN's payment cycle (up to 30 days).

Design implication: The **PayoutRequest** approval step in [payout-design.md](#) should include a check that the earnings being paid out have been covered by a received AWIN payment. This requires LYKE to track AWIN payment receipts.

Simplest implementation: add a **AwinPaymentCoveredAt** date to **CreatorEarning** (or a separate **AwinPaymentBatch** entity), and only allow **Confirmed** earnings that are past **AwinPaymentCoveredAt** to be included in a creator payout request.

For MVP with a single retailer, this can be managed manually by the admin: the "Approve payout" action in the admin UI already exists; the admin approves payouts only after AWIN payment lands in LYKE's bank.

5.2 Transaction Declines After Creator Has Been Paid

If a transaction is declined by AWIN after the creator has already been paid (**EarningStatus** = **Paid**), LYKE faces a shortfall. The network reverses LYKE's commission; LYKE has already paid the creator.

This is the chargeback problem noted as out of scope in [payout-design.md](#). Options:

1. **Clawback** — deduct the reversed amount from the creator's next payout. Requires a **Clawback** earning type and careful UI messaging.
2. **Absorb** — LYKE absorbs small reversals as a cost of doing business (simpler, creates a cap risk at scale)
3. **Reserve fund** — hold back a % of each payout as a rolling reserve against reversals (common in payments — e.g., hold 5% for 90 days)

Recommendation for MVP: Absorb reversals on amounts under £10. For amounts above £10, clawback from next payout. Implement a **Clawback** earning type (negative **Amount**) on the **CreatorEarning** entity.

5.3 Commission Amount Is What LYKE Receives

The **CommissionAmount** in AWIN's postback and API is the **publisher's net commission** — i.e., what LYKE receives from AWIN after AWIN deducts its own network fee. No additional calculation is needed.

The split is then applied on top of this amount:


```
LYKE receives from AWIN: £8.00
Creator share (70%):      £5.60 → CreatorEarning.Amount
LYKE retains (30%):      £2.40 → LYKE platform revenue
```

The `CommissionRate` field in `AffiliateConfig` (currently used informatively) does not need to be used in the calculation — AWIN always reports the actual publisher commission amount.

5.4 Attribution Window Alignment

The `PayoutSettings.AttributionWindowDays` (30 days, used by the old time-based confirmation job) must be updated to reflect that confirmation is now driven by AWIN status, not elapsed time. The field can be kept for display purposes (showing creators an estimated confirmation date) but must no longer gate the `Pending → Confirmed` transition.

Estimated confirmation date shown in the creator earnings UI:

```
Estimated confirmation: {ClickEvent.CreatedAt +
retailer.AffiliateConfig.CookieWindowDays}
```

5.5 Currency

AWIN pays in the retailer's local currency. UK retailers pay in GBP; EU retailers in EUR. AWIN's postback `currency` field specifies the commission currency.

`CreatorEarning.Currency` already stores the currency. `CreatorSettings.DefaultCurrency` is currently `GBP`. If LYKE expands to EU retailers, creator payouts in non-GBP currencies will need to be handled. For MVP (single UK retailer), assume GBP throughout and no FX conversion is needed.

Part 6: Data Model Changes

All changes are additive to the existing schema. No breaking changes.

`Retailer.AffiliateConfig` JSON — add `NetworkType` and `MerchantId`

```
{
  "Enabled": true,
  "NetworkType": "AWIN",          // new: "Direct" | "AWIN" | "Rakuten" |
  "Impact"
  "MerchantId": "7890",          // new: network-assigned merchant/advertiser
  ID
  "BaseUrl": "https://www.awin1.com/awclick.php?mid=7890&id=123456",
  "Parameters": {
    "SubIdParam": "clickref",
    "ProductUrlParam": "p"
  },
  "CommissionRate": 0.08,
```

```

    "CookieWindowDays": 30,
    "WebhookSecret": null           // unused for AWIN; keep for Direct
    integrations
  }

```

ClickEvent.AttributionData JSON — add `awinTransactionId`

```

{
  "source": "feed",
  "creatorId": "...",
  "awinTransactionId": "TXN_abc123",    // new: populated when AWIN postback
  received
  "awinMerchantId": "7890",             // new: for reconciliation
  ...
}

```

New: `EarningStatus.Reversed`

```

public enum EarningStatus
{
    Pending    = 0,
    Confirmed  = 1,
    Paid       = 2,
    Reversed   = 3    // new: AWIN declined/deleted
}

```

New: `EarningType.Clawback` (if absorb threshold implemented)

```

public enum EarningType
{
    Affiliate   = 0,
    Sponsored   = 1,
    Clawback    = 2    // new: negative earning to recover reversed paid
    commissions
}

```

`AffiliateConfig` internal class (in `CommerceService.cs`) — add `NetworkType` and `MerchantId`

```

internal class AffiliateConfig
{
    public bool Enabled { get; set; }
    public string NetworkType { get; set; } = "Direct";    // new
    public string? MerchantId { get; set; }                // new
}

```

```
public string? BaseUrl { get; set; }
public AffiliateParameters? Parameters { get; set; }
public decimal CommissionRate { get; set; }
public int CookieWindowDays { get; set; }
public string? WebhookSecret { get; set; }
public Dictionary<string, string>? CustomParams { get; set; }
}
```

Part 7: New Configuration

AwinSettings

```
public class AwinSettings
{
    public const string SectionName = "Awin";

    // LYKE's AWIN Publisher ID
    public string PublisherId { get; set; } = string.Empty;

    // AWIN API access token (OAuth2 bearer)
    public string ApiToken { get; set; } = string.Empty;

    // AWIN's postback source IP ranges (for allowlisting)
    // Maintain as a comma-separated list; AWIN publishes these in their
    docs
    public string AllowedIpRanges { get; set; } = string.Empty;

    // AWIN API base URL
    public string ApiBaseUrl { get; set; } = "https://api.awin.com";

    // Clawback threshold: reversals above this amount will be clawed back;
    below are absorbed
    public decimal ClawbackThreshold { get; set; } = 10.00m;
}
```

Part 8: New Services and Endpoints

IAwinService (Application layer)

```
public interface IAwinService
{
    // Called by the AWIN postback endpoint
    Task ProcessPostbackAsync(AwinPostbackRequest request,
        CancellationToken ct = default);

    // Called by the reconciliation job
    Task<int> ReconcileTransactionsAsync(DateTime fromDate, DateTime
```

```
toDate, CancellationToken ct = default));

    // Handle a reversal: reverse earning, optionally create clawback
    Task HandleReversalAsync(string awinTransactionId, CancellationToken ct
= default);
}
```

New endpoint: **POST /api/webhooks/awin**

```
POST /api/webhooks/awin
Query params: transaction_id, order_ref, pence, clickref, merchant_id,
status, currency
Auth: IP allowlist (no signature)
Returns: 200 OK always (AWIN does not retry on non-200)
```

Notes:

- Respond **200 OK** immediately regardless of processing outcome. Log errors internally. If LYKE returns non-200, AWIN may retry or stop sending postbacks.
- The endpoint is idempotent: processing the same **transaction_id** twice must be safe (check if earning already exists before creating).

New background job: **AwinReconciliationFunction**

```
Trigger: Timer, daily at 06:00 UTC
Input: AWIN transactions API for the past 48 hours
Action: Sync AWIN transaction statuses to CreatorEarning records
Replaces: EarningConfirmationFunction (time-based logic)
```

Part 9: What Changes vs What Stays the Same

Component	Change needed?	Notes
GenerateAffiliateUrl()	No	Works as-is; just needs correct AffiliateConfig data
ClickEvent tracking	Minor	Add awinTransactionId to AttributionData JSON when postback received
POST /api/retailers/{id}/conversions	Retain	Keep for direct retailer integrations
POST /api/webhooks/awin	New	AWIN-specific postback handler

Component	Change needed?	Notes
<code>EarningConfirmationJob</code>	Replace	Time-based logic removed; replaced by <code>AwinReconciliationFunction</code>
<code>CreatorEarning</code> creation logic	Minor	Same logic, different trigger (AWIN postback vs retailer webhook)
<code>EarningStatus</code> enum	Add value	Add <code>Reversed = 3</code>
<code>EarningType</code> enum	Add value	Add <code>Clawback = 2</code> (if absorbing threshold implemented)
<code>AffiliateConfig</code> internal class	Add fields	<code>NetworkType</code> , <code>MerchantId</code>
Creator payout flow	Policy addition	Admin should only approve payouts after AWIN payment received
Creator earnings UI	Minor	Show estimated confirmation date based on <code>CookieWindowDays</code>
Retailer onboarding in LYKE admin	Minor	Admin sets <code>AffiliateConfig</code> fields when adding a retailer

Part 10: Rollout Sequence

1. **Commercial** — Register LYKE as AWIN publisher. Apply to first retailer's programme. (No code needed yet.)
2. **Config** — Add `AwinSettings` to `appsettings.json`. Populate first retailer's `AffiliateConfig` with AWIN values via admin.
3. **Link generation** — Verify AWIN tracking URLs are generated correctly for the first retailer. Test with a real click in staging.
4. **Postback endpoint** — Deploy `POST /api/webhooks/awin`. Register with AWIN publisher dashboard. Validate postback receipt with a test transaction in AWIN's publisher UI.
5. **Reconciliation job** — Deploy `AwinReconciliationFunction`. Run manually against the previous 7 days to confirm AWIN API connectivity.
6. **Disable time-based confirmation** — Remove or disable `EarningConfirmationFunction`. AWIN is now the source of truth for confirmation.
7. **Reversal handling** — Confirm `Reversed` status transitions work end-to-end with a declined test transaction.
8. **Creator payout policy** — Update admin documentation (or add UI hint) to indicate payouts should only be approved after AWIN settlement is received.

Open Questions

1. **Multiple networks simultaneously** — if LYKE adds Rakuten later, the `IAwinService` should become an `IAffiliateNetworkService` with network-specific implementations. The postback endpoint should be network-agnostic or per-network (`/api/webhooks/awin`, `/api/webhooks/rakuten`). Design for this from the start with `NetworkType` in `AffiliateConfig`.
2. **AWIN IP allowlist maintenance** — AWIN occasionally updates their postback IP ranges. Storing these in config is operational overhead; an alternative is to use a shared secret query parameter (AWIN supports a custom `customref` field that can carry a token).
3. **Reconciliation gap** — if a creator posts content linking to a retailer not yet approved on AWIN, `GenerateAffiliateUrl()` falls back to a direct product URL (existing behaviour). The click is tracked but conversion cannot be attributed. Admin should ensure retailer programme approval before creator content goes live.
4. **AWIN cookie vs app attribution** — AWIN's tracking cookie is set in the browser when the user follows the link from the in-app browser. If the user purchases on a different device, AWIN may not credit the conversion. This is a platform-level limitation of affiliate cookie tracking on mobile. Server-to-server app attribution (AWIN's mobile SDK or fingerprinting) is a potential future enhancement but out of scope for MVP.